

- Q.30 Explain concept of pure electric vehicle. (CO7)
 Q.31 Describe high rate discharging test of battery. (CO2)
 Q.32 Enlist various sensors used in modern vehicle. (CO5)
 Q.33 Describe working of battery ignition system. (CO3)
 Q.34 Enlist five battery defects of lead acid battery. (CO2)
 Q.35 Describe working of mutual switch used in starting system. (CO3)

SECTION-D

- Note:** Long answer type questions. Attempt any two questions out of three questions. (2x10=20)
 Q.36 Explain principle, working and construction of Nickel cadmium battery in detail. (CO3)
 Q.37 Draw a neat sketch of wiring diagram of typical car. (CO2)
 Q.38 Explain various battery rating in detail. (CO3)

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5th Sem / Auto

Subject:- Auto Electrical and Electronics Systems

Time : 3Hrs.

M.M. : 100

SECTION-A

Note: Multiple choice questions. All questions are compulsory (10x1=10)

- Q.1 Instrument used to check Specific gravity of lead acid battery is
 a) Multimeter b) Hydrometer
 c) Hygrometer d) Viscosity Meter
- Q.2 Electrolyte used for alkaline battery is
 a) KOH b) HCL
 c) H₂SO₄ d) NaCl
- Q.3 Tachometer is used for measurement of
 a) RPM b) Distance
 c) Speed d) Fuel consumption
- Q.4 At lower temperature the capacity of battery
 a) Increases b) Decreases
 c) Remains same d) None of these
- Q.5 Number of cells in 6 V battery are
 a) 1 b) 2
 c) 3 d) 6

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- Q.6 The increase in area of plates in battery will
 b) Increase Voltage
 c) Increase Ampere Hour
 d) None
- Q.7 Alternator generates
 a) Direct current b) Alternating Current
 c) Both A & B d) Power
- Q.8 Ignition coil is used to
 a) Step up current b) Step up voltage
 c) Step up current d) Step up power
- Q.9 Chemical energy is converted to _____ energy by battery
 a) Solar b) Potential
 c) Mechanical d) Electrical
- Q.10 The function of dynamo is
 a) Measure the current b) Discharge battery
 c) Charge battery d) Measure voltage

SECTION-B

Note: Objective type questions. All questions are compulsory. (10x1=10)

- Q.11 Write full form of LED.
- Q.12 In India positive earthing is adopted in automobile. (True/False)
- Q.13 Write full form of LT cable.
- Q.14 Write function of wiper motor.

- Q.15 Write use of brake light.
- Q.16 Regulator is the part of _____ system. (charging/Starting/Ignition)
- Q.17 Define sensor.
- Q.18 Write full form of IRVM.
- Q.19 Write material for negative plate for lead acid battery.
- Q.20 Lithium ion batteries are mostly used in electrical vehicles. (T/F)

SECTION-C

Note: Short answer type questions. Attempt any twelve questions out of fifteen questions. (12x5=60)

- Q.21 Enlist and write function of various electrical system of automobile. (CO3)
- Q.22 Describe discharging reaction of lead acid battery. (CO5)
- Q.23 Draw circuit of starting system. (CO1)
- Q.24 Write function of indicating lights and reverse light. (CO5)
- Q.25 Explain how voltage regulation done in charging system? (CO6)
- Q.26 Explain wiring diagram of a two wheeler.
- Q.27 Explain color coding in wiring system of car. (CO3)
- Q.28 Explain concept of parking sensor. (CO6)
- Q.29 Write a short note on battery charging. (CO3)